

FRANCESCO MORI

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PROFESSIONAL SUMMARY

I am a theoretical physicist at Harvard University, supported by the Center of Mathematical Sciences and Applications and a Shuman Educational Research Grant. My research focuses on the dynamics and the control of nonequilibrium systems, with applications to active matter and machine learning.

RESEARCH EXPERIENCE

Research Associate (independent postdoctoral position) Sept. 2025 - Present
Center of Mathematical Sciences and Applications, Harvard University.

Leverhulme-Peierls Fellow (independent postdoctoral position) Oct. 2022 - Sept. 2025
Rudolf Peierls Centre for Theoretical Physics, Department of Physics, University of Oxford.

Junior Research Fellow, New College, Oxford. Oct. 2022 - Sept. 2025

Part-time Consultant, Scroll Prize, Inc. Sept. 2024 - June 2025
Contributing to the [Vesuvius Challenge](#). Image reconstruction of ancient papyri (pre-79 AD).

Ph.D. in Theoretical Physics, Université Paris-Saclay Oct. 2019 - June 2022
Laboratory of Theoretical Physics and Statistical Models (LPTMS), Orsay.
Supervisor: Satya Majumdar.
Title: *Extreme value statistics of stochastic processes: from Brownian motion to active particles.*

TEACHING

Abilitazione Scientifica Nazionale 2025
Accredited to hold Associate Professor positions in Italian universities. (Sections 02/A2 and 02/B2)

Qualification aux fonctions de maître de conférences 2024
Accredited to hold lecturer positions in French universities. (Section 28 - Theoretical Physics)

Stipendiary Lecturer, New College (Oxford) 2023
Mathematical Methods, Thermal Physics.

Tutor, Oxford Study Abroad Program 2023
Biological Physics.

Teaching Assistant, Université Paris-Saclay 2021 - 2022
Computer Science, Statistical Physics.

AWARDS

European Physical Society - Statistical and Nonlinear Physics Division Early Career Prize 2025
for outstanding contributions on the statistical properties of run and tumble particles, thermodynamic cost of stochastic resetting and optimal control protocols.

Shuman Educational Research Grant *9-month research grant.* 2025
Award converted from a Fulbright Scholarship that I was unable to take up due to visa constraints.

EDUCATION

M. Sc. in Physics of Complex Systems, Université Paris-Saclay Sept. 2018 - Jul. 2019
Ranking: 1/42, GPA: 18.6/20

M. Sc. in Physics of Complex Systems, Politecnico di Torino Oct. 2017 - Jul. 2019

GPA: 30.00/30, Final mark: 110/110 cum laude.

M. Sc. in Engineering Physics, Politecnico di Milano
Final mark: 110/110 cum laude.

Oct. 2017 - Jul. 2019

B. Sc. in Applied Mathematics, Politecnico di Torino
GPA: 29.29/30, Final mark: 110/110 cum laude.

Oct. 2014 - Jul. 2017

INTERNSHIPS AND VISITING POSITIONS

Research Visitor, City University of New York.

Feb. 2025 - Present

Research Intern, LPTMS, Orsay (with Satya Majumdar).

Mar. 2019 - Jun. 2019

iMat Project (Project on natural language processing and materials science)
European Materials Modelling Council, Alta Scuola Politecnica.

Jun. 2018 - Sept. 2019

Visiting Student, SISSA and ICTP (Trieste, Italy).

Sept. 2017 - Feb. 2018

Visiting Student, Lund University (Sweden).

Aug. 2016 - Feb. 2017

SCHOLARSHIPS AND TRAVEL GRANTS

Lockey Fund Award (USA) Travel award to attend scientific conferences in the USA.

2024

Lockey Fund Award (Europe) Travel award to attend scientific conferences in Europe.

2024

Astor Travel Scholarship Travel fund for visits to the USA.

2024

Université Paris-Saclay International Master's Scholarship

2018

One-year master's program at Université Paris-Saclay.

Erasmus Scholarship

2018

6-month exchange program at Paris-Saclay University.

Alta Scuola Politecnica

2017

Educational program for the top 1% of master students at Politecnico di Torino and Milano.

Physics of Complex Systems Travel Grant

2017

6-month exchange program at SISSA and ICTP (Trieste, Italy).

Young Talent Project Travel Grant

2016

6-month exchange program at Lund University (Sweden).

Young Talent Project

2014

Educational program for the top 5% of bachelor's students at Politecnico di Torino.

MENTORSHIP

Yaprak Onder (Oxford undergraduate)

2023

Currently Ph.D. student at Harvard University.

Costantino Di Bello (Université Paris-Saclay master's)

2021

Currently postdoctoral researcher at SISSA (Trieste).

This internship resulted in the publication Phys. Rev. E **108**, 014112 (2023).

Marco Biroli (École normale supérieure de Paris master's)

2021

Currently postdoctoral fellow at UChicago.

This internship resulted in the publication J. Phys. A **55**, 244001 (2022).

ACADEMIC SERVICE AND OUTREACH

Organizer, [Harvard CMSA Workshop - Mathematics of Behavior](#).

2026

Organizer , Harvard CMSA Colloquia	2025
Volunteer , Squishy Science Sunday (outreach activity at the APS March Meeting)	Mar. 2025
Assessor for master's project Oxford Interdisciplinary Bioscience DTP	Apr. 2024
Reviewer NeurIPS, SciPost, Cambridge Univ. Press, Nat. Commun., PRL, PRE, J. Phys. A: Math. Theor., J. Stat. Mech., Physica A.	Mar. 2021 - Present
Interviewer , University College (Oxford) Undergraduate Physics admissions	Dec. 2022
Organizer , Cross-TP discussions Journal club across all areas of Theoretical Physics in Oxford	Oct. 2022 - Mar. 2023
Organizer , Fête de la science (outreach activity for high-school students)	Oct. 2021

PUBLICATIONS (* KEY PAPERS)

26. (*) F. Mignacco and **F. Mori**, "A statistical physics framework for optimal learning", PNAS Nexus (2026).
25. (*) B. Bordelon and **F. Mori**, "Theory of Optimal Learning Rate Schedules and Scaling Laws for a Random Feature Model", preprint arXiv:2602.04774 (2026).
24. R. J. Ewart, P. Reichherzer, S. Ren, S. Majeski, **F. Mori**, M. L. Nastac, A. F. A. Bott, M. W. Kunz, A. A. Schekochihin, "Cosmic-ray transport in inhomogeneous media", Mon. Not. R. Astron. Soc. **545**, 2 (2026).
23. (*) **F. Mori** and F. Mignacco. "Analytic theory of dropout regularization", Phys. Rev. E **112**, 045301 (2025). **Featured in Physics.**
22. (*) **F. Mori**, S. Sarao Mannelli, and F. Mignacco. "Optimal Protocols for Continual Learning via Statistical Physics and Control Theory," ICLR 2025 and J. Stat. Mech. 084004 (2025).
 - Accepted for poster presentation at the NeurIPS 2024 workshop Mathematics of Modern Machine Learning.
 - Accepted for poster presentation at COSYNE 2025.
21. (*) **F. Mori** and L. Mahadevan, "Optimal switching strategies for navigation in stochastic settings", J. R. Soc. Interface **22** (227), 20240677 (2025).
20. **F. Mori**, S. N. Majumdar, and P. Vivo. "Cost of excursions until first crossing of the origin for random walk and Lévy flights: An exact general formula", Phys. Rev. Research **6**, 043053 (2024).
19. K. S. Olsen, D. Gupta, **F. Mori**, S. Krishnamurthy, "Thermodynamic cost of finite-time stochastic resetting", Phys. Rev. Research **6**, 033343 (2024).
18. A. Mummery, **F. Mori**, and S. Balbus, "The dynamics of accretion flows near to the innermost stable circular orbit", Mon. Not. R. Astron. Soc. **529**, 1900 (2024).
17. (*) **F. Mori**, S. Bhattacharyya, J. M. Yeomans, and S. P. Thampi, "Viscoelastic confinement induces periodic flow reversals in active nematics", Phys. Rev. E **108**, 064611 (2023).
16. S. N. Majumdar, **F. Mori**, and P. Vivo, "Nonlinear-Cost Random Walk: exact statistics of the distance covered for fixed budget", Phys. Rev. E **108** (6), 064122 (2023).
15. C. Di Bello, A. K. Hartmann, S. N. Majumdar, **F. Mori**, A. Rosso, and G. Schehr, "Current fluctuations in stochastically resetting particle systems", Phys. Rev. E **108**, 014112 (2023). **Highlighted as an Editors'**

Suggestion.

14. S. N. Majumdar, **F. Mori**, and P. Vivo, "The cost of diffusion: nonlinearity and giant fluctuations", **Phys. Rev. Lett.** **130**, 237102 (2023).
13. (*) B. De Bruyne and **F. Mori**, "Resetting in Stochastic Optimal Control", *Phys. Rev. Research* **5**, 013122 (2023).
12. (*) **F. Mori**, K. S. Olsen, and S. Krishnamurthy, "Entropy production of resetting processes", *Phys. Rev. Res.* **5**, 023103 (2023).
11. **F. Mori**, S. N. Majumdar, and G. Schehr, "Time to reach the maximum for a stationary stochastic process", *Phys. Rev. E* **106**, 054110 (2022).
10. M. Biroli, **F. Mori**, and S. N. Majumdar, "Number of distinct sites visited by a resetting random walker", *J. Phys. A: Math. Theor.* **55**, 244001 (2022).
9. **F. Mori**, G. Gradenigo, and S. N. Majumdar, "First-order condensation transition in the position distribution of a run-and-tumble particle in one dimension", *J. Stat. Mech.* 103208 (2021).
8. (*) **F. Mori**, S. N. Majumdar, and G. Schehr, "Distribution of the time of the maximum for stationary processes", *Europhys. Lett.* **135**, 30003 (2021). **Highlighted as an Editors' Choice.**
7. **F. Mori**, P. Le Doussal, S. N. Majumdar, and G. Schehr, "Condensation transition in the late-time position of a run-and-tumble particle", *Phys. Rev. E* **103**, 062134 (2021).
6. S. N. Majumdar, **F. Mori**, H. Schawe, and G. Schehr, "Mean perimeter and area of the convex hull of a planar Brownian motion in the presence of resetting", *Phys. Rev. E* **103**, 022135 (2021).
5. **F. Mori**, P. Le Doussal, S. N. Majumdar, and G. Schehr, "Universal properties of a run-and-tumble particle in arbitrary dimension", *Phys. Rev. E* **102**, 042133 (2020). **Highlighted as an Editors' Suggestion.**
4. B. Lacroix-A-Chez-Toine, **F. Mori**, "Universal survival probability for a correlated random walk and applications to records", *J. Phys. A: Math. Theor.* **53**, 495002 (2020).
3. (*) **F. Mori**, P. Le Doussal, S. N. Majumdar, and G. Schehr, "Universal survival probability for a d -dimensional run-and-tumble particle", **Phys. Rev. Lett.** **124**, 090603 (2020).
2. **F. Mori**, S. N. Majumdar, and G. Schehr, "Distribution of the time between maximum and minimum of random walks", *Phys. Rev. E* **101**, 052111 (2020).
1. **F. Mori**, S. N. Majumdar, and G. Schehr, "Time between the maximum and the minimum of a stochastic process", **Phys. Rev. Lett.** **123**, 200201 (2019).

INVITED TALKS

Statistical Physics & Machine Learning: moving forward Institut d'Études Scientifiques de Cargèse (France).	2025
Paris Biological Physics Community Day École normale supérieure (Paris).	2024
Workshop: Stochastic Systems in Active Matter Isaac Newton Institute (Cambridge).	2024
Workshop: New Vistas in Stochastic Resetting The Higgs Centre for Theoretical Physics (Edinburgh).	2024
Saturday Mornings of Theoretical Physics (outreach activity for Oxford Physics alumni) University of Oxford.	2023
Theoretical Physics Colloquium University of Oxford.	2022

Large Deviations, Extremes and Anomalous Transport in Non-equilibrium Systems The Erwin Schrödinger International Institute for Mathematics and Physics (Austria).	2022
Nordita Scientific Program “Are there universal laws in nonequilibrium physics” Nordita Institute, Stockholm (Sweden).	2022

INVITED SEMINARS

Harvard Soft Condensed Matter Seminar Harvard University, Department of Physics.	2026
MIT Physics of Living Systems Seminar Massachusetts Institute of Technology.	2026
LPTMC seminars Laboratoire de Physique Théorique de la Matière Condensée, Paris.	2022, 2023, 2024, and 2026
LPTMS Seminar Laboratory of Theoretical Physics and Statistical Models, Orsay.	2026
MIT Physics of Living Systems Short Talk Massachusetts Institute of Technology	2025
Mathematical Physics Group Seminar University of Bologna.	2025
Disordered System Seminar King's College London.	2022 and 2025
Condensed Matter Seminar University of Massachusetts Amherst.	2025
Soft Matter Seminar University of California, Berkeley.	2025
ML Nosh Lunch Seminar University of Oxford.	2025
Soft Matter Group Away Day University of Oxford.	2024
Soft Matter Seminar University of California, Santa Barbara.	2023
Soft Condensed Matter Seminar New York University.	2023
IPhT Seminar Institut de Physique Théorique, Saclay.	2023
LOMA Seminar Laboratoire Ondes et Matière d'Aquitaine, Bordeaux.	2023
Statistical Physics and Complexity Webinar Series University of Edinburgh.	2022
LuxStatMech seminar University of Luxembourg.	2022
SIFS Young Seminar Italian Society of Statistical Physics.	2022

ICTS Statistical Physics Journal Club

International Centre for Theoretical Sciences, Bangalore.

2021

CONTRIBUTED TALKS

Journée “Physique et Vivant”

Institut Jacques Monod (Paris).

2023

Nordita Workshop: Fluctuations and First-Passage Problems

Nordita Institute, Stockholm (Sweden).

2023

4th Course on Multiscale Integration in Biological Systems

Institut Curie, Paris (France).

2021

Journée Systèmes & Matière Complexes

Université Paris-Saclay, Paris (France).

2021